

Erdős Problem #776

EP-776

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $r \geq 2$ and $A_1, \dots, A_m \subseteq \{1, \dots, n\}$ be such that $A_i \not\subseteq A_j$ for all $i \neq j$ and for any t if there exists some i with $|A_i| = t$ then there must exist at least r sets of that size.

How large must n be (as a function of r) to ensure that there is such a family which achieves $n - 3$ distinct sizes of sets?